

Student Name: Singh's Solutions
+ marking guide
2013.

Teacher's Name: _____

Total marks = [/ 45] = _____ %

Section A: Ten multiple choice questions. Answer **ALL** the questions by writing in the correct answer **neatly and clearly** in the answer table below. One mark for each correct answer. Read the questions carefully. [10 marks]

M/C: 1 mark for each correct Answer.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
D	B	C	A	D	A	C	D	C	B

Q1. In which of the following pairs is the atomic radius of the second atom greater than the first atom?

- (A) Be, B
 (B) B, F
 (C) Cl, Ar
 → ~~(D) Ar, K~~

Q2. Which of the following species does **not** have an electron configuration the same as a group 18 element.

- (A) Ne
 → ~~(B) He²⁺~~
 (C) Cl⁻¹
 (D) Mg²⁺

Q3. Which one of the following substances is composed of positive and negative ions?

- (A) Fe
 (B) CO₂ (g)
 → ~~(C) CaO~~
 (D) HCl (g)

Q4. Which of the following correctly lists the molecules in order of increasing polarity?



least polar ----- most polar

-  ~~(A)~~ $\text{O}_2 < \text{CO}_2 < \text{HBr} < \text{HCl}$
- (B) $\text{HBr} < \text{HCl} < \text{CO}_2 < \text{O}_2$
- (C) $\text{O}_2 < \text{CO}_2 < \text{HCl} < \text{HBr}$
- (D) $\text{HCl} < \text{HBr} < \text{O}_2 < \text{CO}_2$

Q5. Which alternative correctly shows the particles present in solid samples of carbon dioxide, silicon dioxide and magnesium oxide

carbon dioxide

silicon dioxide

magnesium oxide

- | | | | |
|--|-----------|-----------|-----------|
| (A) | molecules | molecules | ions |
| (B) | molecules | ions | ions |
| (C) | atoms | atoms | molecules |
|  (D) | molecules | atoms | ions |

Q6. Which of the following correctly shows the shapes of the respective molecules?

- Let T = trigonal planar : P = pyramidal : L = linear : B = bent

H_2CO

AsH_3

BF_3

H_2S

- | | | | | |
|--|---|---|---|---|
|  (A) | T | P | T | B |
| (B) | T | P | P | B |
| (C) | P | P | T | B |
| (D) | P | T | P | L |

Q7. The molecular substance of formula XY is very soluble in water and only slightly soluble in carbon tetrachloride (CCl_4). What is the major or dominant type of intermolecular force(s) present in XY

- (A) hydrogen bonding
- (B) covalent bonding
-  ~~(C)~~ hydrogen bonding or dipole – dipole forces
- (D) dispersion forces

Q8. Which one of the following species does NOT have eight valence electrons around its central atom?

- (A) CCl_4
- (B) NH_3
- (C) AsH_3
-  (D) ~~(D)~~ BCl_3

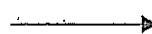
Q9. Which one of the following compounds should have the highest boiling point?

- (A) NH_3
- (B) CH_4
-  (C) ~~(C)~~ H_2O
- (D) HF

Q10. Which one of the following substances contains only ONE non-bonding pair of valence electrons?

- (A) NH_4^+
-  (B) ~~(B)~~ HCN
- (C) C_2H_4
- (D) N_2

- End of section A multiple choice questions.

- Go onto to short answer section B 

SECTION B: SHORT ANSWER SECTION [27 MARKS]

1. Write your answers in the spaces provided below.
2. Marks for each question is written next to it.
3. Answer **ALL** the questions in this section.
4. Note that not all questions carry equal number of marks

Q1. write out the chemical formula and correct electron configuration for:

- (a) magnesium atom = $\text{Mg}^{\frac{1}{2}}$ = 2, 8, 2 $\frac{1}{2}$ } [1 mark] ✓①
- (b) nitride ion = N^{3-} = 2, 8 $\frac{1}{2}$ } [1 mark] ✓①
- (c) phosphide ion = P^{3-} = 2, 8, 8 $\frac{1}{2}$ } [1 mark] ✓①

Q2. Explain the trend in the first ionisation displayed by the group II elements shown below. [2 marks]

(a) Elements: Magnesium Calcium StrontiumFirst ionisation energy: 0.744 0.596 0.556
in MJ mol⁻¹

- 1st I.E. ↓ from Mg → Sr because valence electrons...
entering orbitals are further away in distance (↑)
from the attraction of the nucleus. ✓①
- For each elements full inner shells' electrons
also reduces attractive power of the nucleus
by "electron shielding effect". ✓
- Effective nuclear charge ↓ down the group as atomic/ionic radii ↑ ✓

Q2 (b) Explain the variation in the successive ionisation energies for Lithium [2 marks]

Successive ionisation energy First Second Thirdin MJ mol⁻¹ 0.526 7.30 11.8

- I.E. ↑ dramatically (0.526 MJ) to (7.30 MJ)
indicating an increase in the positive charge
on the Li⁺ ion (3p⁺) ✓①
- Removal of the 2nd electron from the K shell (n=1)
is much closer to the nucleus hence there is a
very strong attraction between +ve nucleus and e⁻
and this causes successive I.E. to increase.

Q3. A student was given four substances and asked to perform a series of tests to determine whether they were metallic, covalent molecular, ionic or covalent network substances. The results obtained from the tests are tabled below.

Substance	Melting point	Electrical conductivity as a solid	Electrical conductivity when molten	Electrical conductivity in aqueous solution	Hardness
A	Liquid at room temp	Not tested	Non conductive	Non conductive	Not tested
B	250°C	conducts	Conducts	Does not dissolve	Will bend
C	2760 °C	Non conductive	Non conductive	Does not dissolve	Very hard
D	1580 °C	Non conductive	Conducts	conducts	Brittle

Now complete the table as follows:

(a) In the bond type column, state whether each substance is metallic (M), covalent molecular (CM), ionic (I) or covalent network (CN). [2 marks]

(b) In the explanation column, explain your decisions using the information in the table. [4 marks]

Substance	Bond type	Explanation
A	CM $\checkmark$	Low m.p. - Liq @ Room Temp $\checkmark$ $\therefore$ weak intermolecular forces. $\checkmark$ Non conductive as there are no charged particles formed. $\checkmark$
B	M $\checkmark$	Charged particles available $\checkmark$ ① in <u>solid</u> and <u>molten</u> (ions) Malleable - due to sea of delocalised electrons between metallic ions. $\checkmark$
C	CN $\checkmark$	High m.p. atoms linked throughout $\checkmark$ the 3D structure by <u>strong covalent bonds</u> . $\checkmark$ No electrically charged carriers available $\checkmark$ $\therefore$ its a non conductor $\checkmark$
D	I $\checkmark$	Strong electrostatic attraction $\checkmark$ between (+ve) and (-ve) ions $\therefore$ high m.p. Brittle due to crystalline structure Ions fixed in solids $\Rightarrow$ non conductor. Conductors in dissolved solutions $\checkmark$ and molten state due to mobile ions present as charge carriers $\checkmark$

Q4. Draw a valid electron dot diagram for ALL the following chemical molecules. Either name or sketch a three dimensional diagram to indicate their shape and polarity. [8 mks]

$\frac{1}{2}$ MK/BOX MAX.	(a) electron dot diagram	(b) name or 3-D sketch of shape ^{only} $\frac{1}{2}$ MK for Name or sketch.	(c) polarity of molecule
(i) phosphine (PH ₃)		PYRAMIDAL	POLAR $\frac{1}{2}$
(ii) methane (CH ₄)		Tetrahedral	NON POLAR $\frac{1}{2}$
(iii) water (H ₂ O)		'V' shaped or Bent	POLAR $\frac{1}{2}$
(iv) dichloromethane (CH ₂ Cl ₂)		Tetrahedral	POLAR $\frac{1}{2}$

Q5. In each of the following pairs, **circle** which substance you would expect to have greater solubility in water. Give reasons for your choices briefly. [6 marks]

(a) HCN or N₂

Reasons for choice: HCN more soluble than N₂ gas in H₂O.

HCN is polar due to dipole-dipole intermolecular forces.
N₂ has weak dispersion forces hence unable to break strong H-bonding in water.

(b) C₂H₆ or NaCl

Reasons for choice: C₂H₆ only weak dispersion forces between molecules. Whereas Na⁺Cl⁻ strong ionic bonding (extreme polar) easily able to dissolve in water molecules (H-bonding).

(c) NH₃ or HCl

Reasons for choice: NH₃ has strong H-bonding between molecules.

Hence bonds easily with polar H-bonded H₂O molecules
HCl less soluble due to weaker dipole-dipole forces
 End of section B: Go to section C: Problem solving and calculations
 intermolecular forces.

SECTION C: Problem solving and calculations [8 marks total]

1. Show working out steps carefully to score the full marks
2. Numerical answers should be stated correct to three significant figures

Q1. A solution of ammonium sulfate was prepared by dissolving 7.35 g of $(\text{NH}_4)_2\text{SO}_4$ in a small volume of distilled water and the resultant solution was made up to 250.0 mL using a volumetric flask. The final solution had a mass of 256.7 g. Determine:

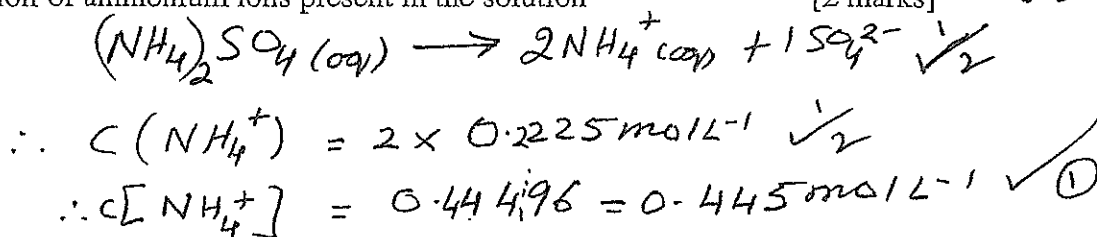
- (a) concentration of ammonium sulfate $(\text{NH}_4)_2\text{SO}_4$ solution in mol L^{-1} [2 marks]

$$n[(\text{NH}_4)_2\text{SO}_4] = \frac{m}{M} = \frac{7.35\text{g}}{132.144} \checkmark \quad M[(\text{NH}_4)_2\text{SO}_4] = (2 \times 14.01) + (8 \times 1.008) + 32.06 = 132.144\text{g mol}^{-1} \checkmark$$

$$\therefore C[(\text{NH}_4)_2\text{SO}_4]_{\text{solution}} = \frac{0.05562\text{ mol}}{V(0.250)} = 0.2225\text{ mol L}^{-1} \checkmark$$

$$\therefore \text{conc}^n(\text{E})[(\text{NH}_4)_2\text{SO}_4]_{\text{solution}} = 0.2225\text{ mol L}^{-1} \checkmark$$

- (b) concentration of ammonium ions present in the solution [2 marks]



- (c) concentration of ammonium sulfate solution in grams per litre (g L^{-1}) [2 marks]

method 1 either $\rightarrow C[(\text{NH}_4)_2\text{SO}_4] = \frac{m}{V} = \frac{7.35\text{g}}{0.250} = 29.4\text{ g L}^{-1} \checkmark \text{ ①}$

or $\rightarrow C[(\text{NH}_4)_2\text{SO}_4]_{\text{solution}} = 0.2225\text{ mol L}^{-1} \checkmark \text{ ①}$

method 2 $\rightarrow \therefore C[(\text{NH}_4)_2\text{SO}_4] = 0.2225\text{ mol L}^{-1} \times 132.144\text{ g mol}^{-1} = 29.4\text{ g L}^{-1} \checkmark \text{ ①}$

- (d) concentration of the ammonium sulfate solution in parts per million (p.p.m.) [2 marks]

$$C[\text{p.p.m.}] = \frac{\text{mg solute}}{\text{kg solution}} \checkmark$$

$$m(\text{solution}) = 256.7\text{g} = 0.2567\text{kg} \checkmark$$

$$= \frac{7.35 \times 10^3\text{mg}}{0.2567\text{kg}} = 28632.6\text{ p.p.m.} \checkmark$$

$$\therefore C[(\text{NH}_4)_2\text{SO}_4]_{\text{solution}} = 2.86 \times 10^4\text{ p.p.m.} \checkmark \text{ ①}$$

END OF BONDING TEST 2013: GO BACK AND RE-CHECK YOUR ANSWERS.